

Chapter Two

Data Representations

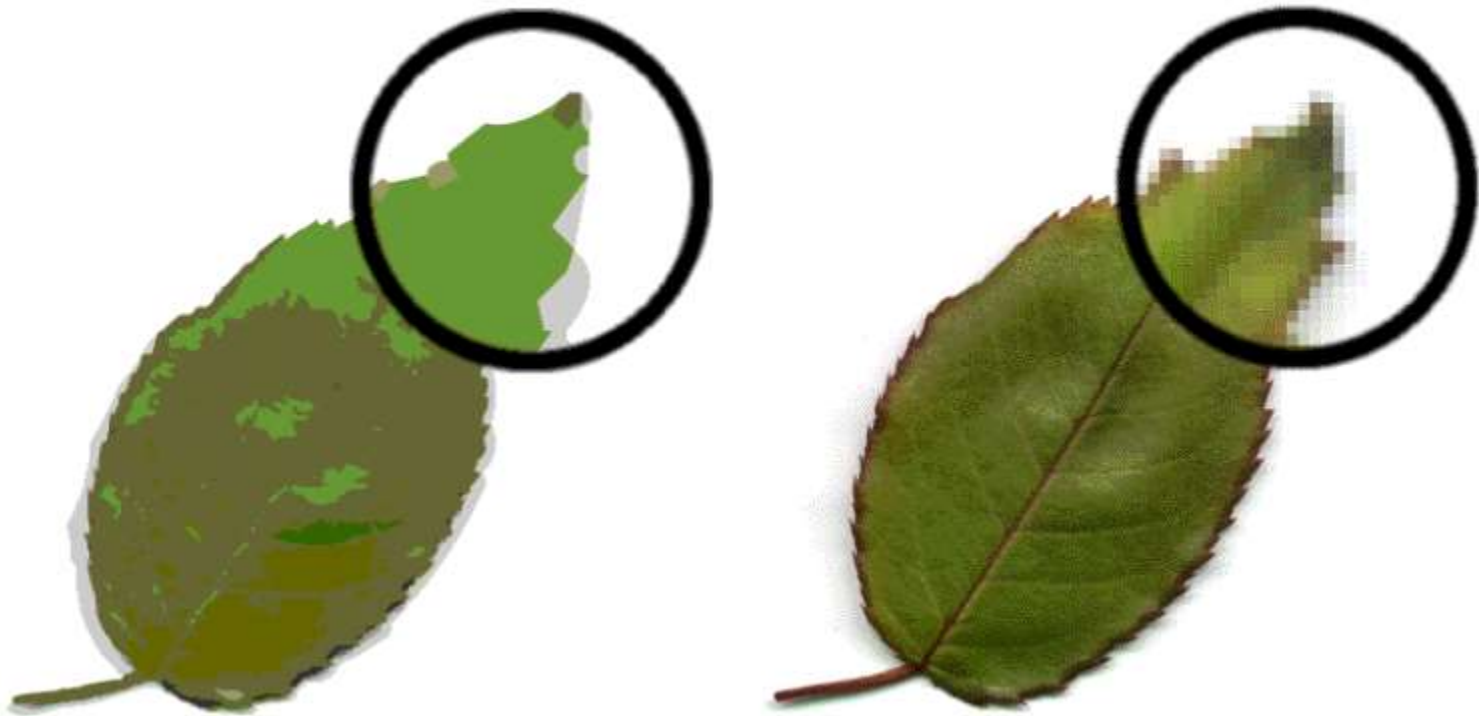
Outline

- graphics /image data representation
- popular file formats

Vector Graphics

- Vector graphics define pictures as collections of lines, ellipses, triangles, polygons and other basic graphic "primitives."
- Vector graphics are limited to storing mechanical art
- Vector graphic formats include
 - Corel Draw files, which use the extension CDR,
 - AutoCAD drawings, which use the extension DXF.

Bitmap vs. Vector graphics



left side----> vector graphics image

Right side----> bit map/raster image

Compression

Graphic file formats can be regarded as being of three types.

- The first type stores graphics **uncompressed**.
 - Windows BMP (.bmp) files are an example of this sort of format.
- The second type is called **"non-lossy"** or **"lossless"** compression formats.
 - Most graphic formats use lossless compression - the GIF formats are among them.
- The third type of bitmapped graphic file formats is called **"lossy"** compression.
 - the details are what prevent areas from being all the same color, and as such from responding well to compression.
 - perhaps too subtle to be discernable by your eye

Analog versus bitmap images

- The visual world is analog, which is to say that real-world images is a continuous spectrum of colors.
- Infinity is uncomfortably large to a computer. A computer-stored image is split into little bits of light.
- A bitmap is a matrix of colored dots.

Bitmaps, Pixels & Colors

- The colored dots that make up a bitmap are properly called "pixels".
- Color is represented on a computer by using varying amounts of red, green and blue light.
- In the simplest sort of bitmapped image, each pixel is represented by three numbers to store the amounts of red, green and blue light that define the color of the pixel in question.

True Color Bitmap

- Each pixel requires one byte for each color index for a total of three bytes per pixel.
- As a byte represents eight bits, each pixel requires 24 bits to store all its color information.
- This defines the maximum number of discrete colors this sort of bitmap can represent as 2^{24} , or 16,777,216.
- Such graphics are referred to as "True Color" images, or just as "24-bit" graphics.

Bit-Depth = Color-Depth

- Number of Colors = $2^{(\text{Bit-depth})}$
- Bit-depth is the number of bits.
 - It is also called “Color resolution”.

Bit depth	Color resolution	Calculation
1-bit	2 colors	$2^1 = 2$
2-bit	4 colors	$2^2 = 4$
3-bit	8 colors	$2^3 = 8$
4-bit	16 colors	$2^4 = 16$
8-bits	256 colors	$2^8 = 256$
16-bits	65,536 colors	$2^{16} = 65536$
24-bits	16,777,215 colors	$2^{24} = 16.7 \text{ million}$

Image File Formats

Examples of common interchange formats are:

- GIF - Graphics Interchange Format
- PNG - Portable Network Graphics
- JFIF - JPEG (Joint Photographic Experts Group) File Interchange Format
- TIFF - Tagged Image File Format
- PGM - Portable Gray Map
- -PBM - Portable Bit Map
- -PPM - Portable Pix Map
- - FITS - Flexible Image Transport System

Test Image in Various Formats

- JPG 67 kb
- GIF 185 kb (256 colors, optimized)
- BMP 899 kb ($640 \times 479 \times 3 / 1024 = 898.125$)
- TIF 902 kb
- PNG 418 kb
- RAW 900 kb

GIF (Graphics Interchange Format)

- GIF creates a table of up to 256 colors from a pool of 16 million.
- If the image has fewer than 256 colors, GIF can render the image exactly.
- GIF is "lossless" only for images with 256 colors or less. For a rich, true color image, GIF may "lose" 99.998% of the colors.
- The GIF format can contain animations. Its internal structure is such that it can store multiple images and the controls to make them appear as real time animation
 - animated GIF.
- The GIF format also allows a special color as to be specified as "using the background." This results in the image looks like transparent
 - transparent GIF.

Animated GIF

- Advantages:
 - No need other software or plugins
 - GIF is the standard format on the Web.
 - GIF animated tools are available to use.
- The file contains layered frames on top of each other.
- Tips:
 - Avoid more than one animated GIF on a page.
 - Avoid animated GIF on text-rich pages.
 - Examine the pause between repetitions.



Transparency GIF

- The image can be shapes other than rectangles.
- One position of the color palette is designated as "Transparent".
- All pixels of the image that have this particular color index will be painted as transparent when viewing.

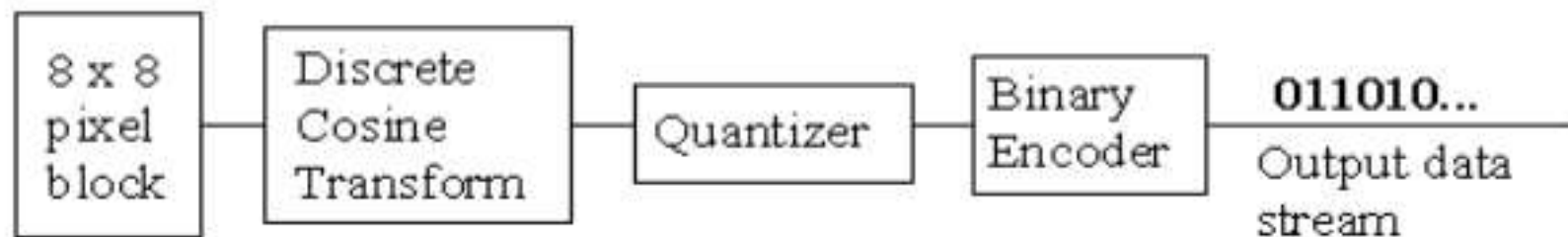


a transparent GIF

JPG/JPEG Format

- JPEG – Joint Photographic Experts Group
- The JPEG format uses "lossy" compression to get more graphics into a smaller file than would otherwise be possible.
- An image written to the JPEG format will be degraded.
- The amount of degradation, the "quality factor," can usually be set in a graphics software.
 - If the value is set to 100, almost no degradation will occur.
 - If it is set to a value close to zero, the resulting image will be a very small file but unrecognizable.
 - The default value of 75 is usually a good compromise.

FIGURE 1. BLOCK DIAGRAM OF JPEG COMPRESSION



Progressive JPEG

- There are two types of JPEG files extant as of this writing, called "sequential" and "progressive".
 - A **sequential JPEG** file stores its image as a simple bitmap.
 - A **progressive JPEG** files stores its image such that it can appear initially out of focus when it begins to download to a web page, and resolve itself as more of the image is received by your web browser.
 - Advantage: Provide indication of the whole image to the viewer before the entire image is loaded.
 - Disadvantage: Require more computational power to display.

GIF Versus JPG

- For photographic images JPG is better both for quality and file size
- For line art GIF is better both for quality and file size
- The future of line art is probably in vector file formats like SVG (Scalable Vector Graphics)
- VML (Vector Markup Language) is Microsoft backed

	GIF	JPEG
Best application	Line Art, Image with few color text	Photographs, Image with many colors
How to reduce display time?	Interlace	Interlace (Progressive)
Display speed	Fast	Slower, more computation
Benefits	Transparency, Animation	Greatest compress for photographs, more color
Max. color	256	16.7 million

TIFF (Tagged Image File Format)

- TIFF can be lossless or lossy.
- The details of the image storage algorithm are included as part of the file.
- TIFF is used almost exclusively as a lossless image storage format that uses no compression at all.
- Most graphics programs that use TIFF do not use compression .
- File sizes are quite big.
 - (Sometimes LZW is used, but it is not universally supported.)

PNG (Portable Network Graphics)

- PNG is a lossless storage format.
- It looks for patterns in the image that it can use to compress file size.
- The compression is exactly reversible, so the image is recovered exactly.
- Support three image type: true color, grayscale, palette-based (8-bit).
 - JPEG supports the first 2.
 - GIF supports the 3rd one.

Advantages

- Better Compression
 - Deflate is an improved version of the Lempel-Ziv compression algorithm.
- Improve Interlacing
 - Display image quicker than Interlaced GIF.
- True Color and Transparency
 - Support 16-bit (Grey scale) or 48-bit (True Color)
 - 16-bit for alpha channel (Transparency).
- Gamma storage
 - Store the gamma setting of the platform of the creator.

Disadvantages

- Not support by old browsers (Netscape 2,3,4 and IE 2,3,4)

BMP bitmap (DIB) file format /bitmap

- **BMP** is an uncompressed proprietary format invented by Microsoft.
- There is really no reason to ever use this format.

PSD

- PSD is a proprietary format used by Photoshop.
- Working format to edit images in the software
- This package uses layers to build complex images, and layer information may be lost in the nonproprietary formats such as TIFF and JPG.
- However, it is best to save the product as a TIFF or JPG, so it can be viewed in the future when software changes.

Number of colors

- Images have differing numbers of color within them.
- Black and white images take up the least amount of space with 1 bit of information.
- Current video cards are able to display 24 bits of color per pixel.
- This makes it possible for 16 million colors to be displayed (2^{24} colors).
- This requires a significant amount of disk space.



8-bit and 24-bit photos

- Can you see the difference?



left= 8-bit
right=24-bit

Color Palettes

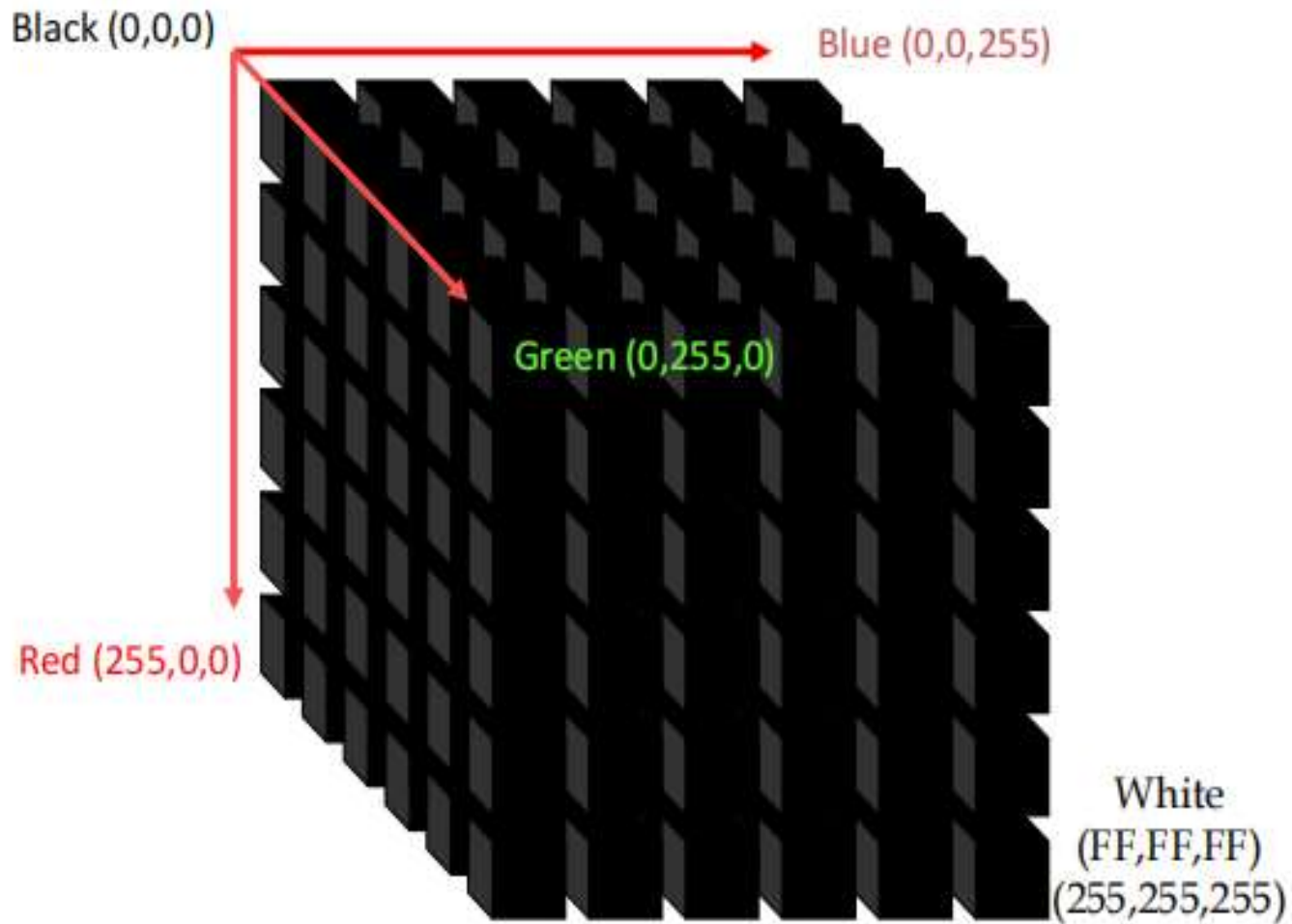
The store images having a maximum of 16,777,216 colours, is one which can store images having a maximum of 256 colours.

- Such files are referred to as using "palette-colour". The colours in a palette-colour file are derived from a potential palette of 16,777,216 colours, but no more than 256 of them can be used in any one image.

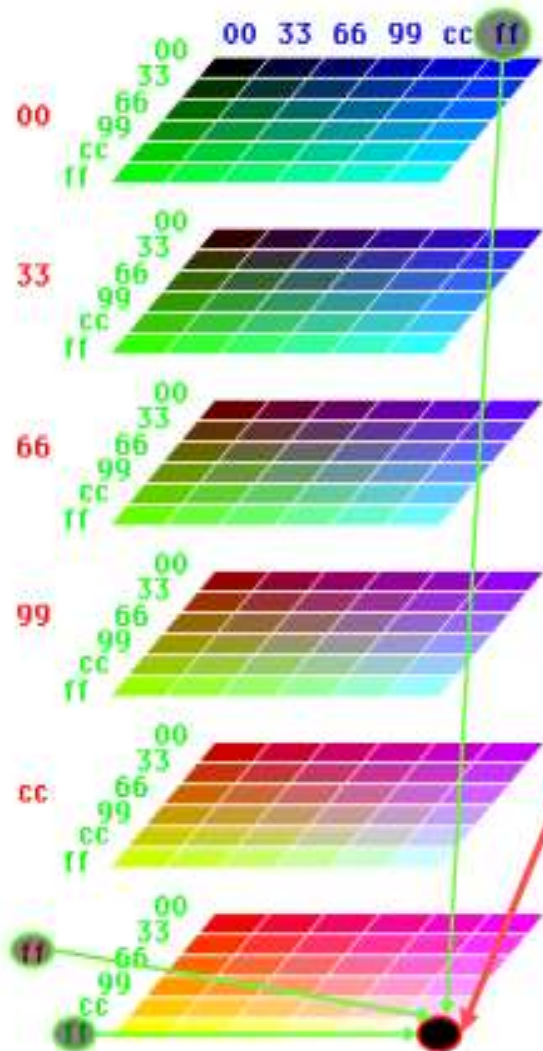
The Web Palette

- Color in Browser – 24-bit color displays on an 8-bit monitor display.
- Need a common palette (Web Palette) for browser, guarantees the image will look the same on different platforms.
- If the browser is in 24-bit display, the palette does not affect the images.
- Web Palette consists of 216 colors
- Also called: Web-safe Palette, Non-dithering Palette, 6x6x6 cube.

Which Color Palette?



6x6x6 Color Palette



Learning Web Design: The Web Palette - Microsoft Internet Explorer

File Edit View Favorites Tools Help

Back Forward Stop Search Favorites Media Print View Source Address http://www.les...

Links Yahoo Mail CSC1720 OldHomepage Dictionary RTM Google Search Web

HTML Color Names

List of Links

Books

255, 255, 255 FFFFFF	255, 255, 204 FFCCFF	255, 255, 153 FFCC99	255, 255, 102 FFCC66	255, 255, 51 FFCC33	255, 255, 0 FFCC00
255, 204, 255 FFCCFF	255, 204, 204 FFCCCC	255, 204, 153 FFCC99	255, 204, 102 FFCC66	255, 204, 51 FFCC33	255, 204, 0 FFCC00
255, 153, 255 FF99FF	255, 153, 204 FF99CC	255, 153, 153 FF9999	255, 153, 102 FF9966	255, 153, 51 FF9933	255, 153, 0 FF9900
255, 102, 255 FF66FF	255, 102, 204 FF66CC	255, 102, 153 FF6699	255, 102, 102 FF6666	255, 102, 51 FF6633	255, 102, 0 FF6600
255, 51, 255 FF33FF	255, 51, 204 FF33CC	255, 51, 153 FF3399	255, 51, 102 FF3366	255, 51, 51 FF3333	255, 51, 0 FF3300
255, 0, 255 FF00FF	255, 0, 204 FF00CC	255, 0, 153 FF0099	255, 0, 102 FF0066	255, 0, 51 FF0033	255, 0, 0 FF0000
204, 255, 255 CCFFFF	204, 255, 204 CCFFCC	204, 255, 153 CCFF99	204, 255, 102 CCFF66	204, 255, 51 CCFF33	204, 255, 0 CCFF00
204, 204, 255 CCCCFF	204, 204, 204 CCCCCC	204, 204, 153 CCCC99	204, 204, 102 CCCC66	204, 204, 51 CCCC33	204, 204, 0 CCCC00
204, 153, 255 CC99FF	204, 153, 204 CC99CC	204, 153, 153 CC9999	204, 153, 102 CC9966	204, 153, 51 CC9933	204, 153, 0 CC9900
204, 102, 255 CC66FF	204, 102, 204 CC66CC	204, 102, 153 CC6699	204, 102, 102 CC6666	204, 102, 51 CC6633	204, 102, 0 CC6600
204, 51, 255 CC33FF	204, 51, 204 CC33CC	204, 51, 153 CC3399	204, 51, 102 CC3366	204, 51, 51 CC3333	204, 51, 0 CC3300
204, 0, 255 CC00FF	204, 0, 204 CC00CC	204, 0, 153 CC0099	204, 0, 102 CC0066	204, 0, 51 CC0033	204, 0, 0 CC0000
153, 255, 255 99FFFF	153, 255, 204 99FFCC	153, 255, 153 99FF99	153, 255, 102 99FF66	153, 255, 51 99FF33	153, 255, 0 99FF00

Done Internet

Define RGB Values

- **Decimal**

- Ranging from 0 to 255.
- 51-51-255 means red value is 51, green value is 51, blue value is 255.

- **Hexadecimal**

- Base-16 number system
- Only six characters to describe RGB color.
- 51-51-255 is 3333FF.

- **Percentage**

- 51-51-51 is 20%-20%-100%
- 00,33,66,99,CC,FF - 0%,20%,40%,60%,80%,100%
- Some old Macintosh programs use it.

Gamma control

- the overall brightness of the display of a computer monitor.
- The relationship of voltage and the light intensity.
- Default gamma setting varies from platform to platform
 - Macintosh – 1.8 Gamma
 - PC – 2.2 Gamma
 - UNIX – 2.4 Gamma
- The higher the gamma value, the darker the display.
- Image created on PC looks wash out on Macintosh.



Different Gamma Value

Thank You !